

Data, Model, and Estimation Design

Competition and Inflation under HSA Demand: A Nine-Cell Identification-First Local State-Space Design

Final Research Design Blueprint

Abstract

This document specifies a pre-declared nine-cell, identification-first semi-structural design for studying whether an empirical business-demography competition coordinate changes inflation dynamics through a competition-dependent marginal-cost or slack slope and a short-run market-participation channel. The nine cells combine three quarterly inflation/expectation blocks—PPI with GDP-deflator expectations, CPI with CPI expectations, and core CPI with CPI expectations—with three activity measures—inverse markup, a Beveridge–Nelson output gap, and a negative unemployment gap.

The cells are not given symmetric structural interpretations. The PPI–inverse-markup cell is the unique quantitative HSA cell and is used to evaluate the local cross-equation restriction

$$\kappa_1 = \zeta_0 \theta_0.$$

The remaining eight cells are pre-declared mechanism-robustness or transportability exercises and do not receive the quantitative HSA equality. Annual firm counts anchor the empirical competition coordinate, while quarterly business-demography indicators provide additional information about its stationary competition cycle. Full-joint and modular-cut analyses are reported side by side. Identification diagnostics are co-primary with model evidence.

The flexible-price/steady-state reference markup is estimated jointly rather than imposed in a preliminary calibration. The same reference markup determines the inverse-markup marginal-cost deviation and the HSA reference elasticity, eliminating sample-mean normalization from the theory-linked cell. Bayes factors measure integrated within-cell model evidence, time-series-respecting predictive scores evaluate predictive performance, and posterior predictive checks diagnose absolute fit. None of these evidence measures overrides weak-identification diagnostics.

Contents

1 Objective and Scope

5

2	Research Questions	5
2.1	RQ1: Competition-Dependent Slope	5
2.2	RQ2: Short-Run Participation Channel	5
2.3	RQ3: Local HSA Restriction	6
2.4	RQ4: State Dependence of the Participation Loading	6
2.5	RQ5: Transportability	6
3	Nine Pre-Declared Empirical Cells	7
3.1	Tier 1: Quantitative HSA Cell	7
3.2	Tier 2: Mechanism Robustness	7
3.3	Tier 3: Macro Transportability	8
4	Inflation and Expectation Measures	8
4.1	PPI Cells	8
4.2	CPI Cells	9
4.3	Core CPI Cells	9
5	Marginal-Cost and Slack Measures	9
5.1	Inverse Markup	9
5.2	BN Output Gap	9
5.3	Negative Unemployment Gap	10
6	HSA Theory and Local Restriction	10
7	Competition Measurement System	11
7.1	Annual Firm Counts	11
7.2	Quarterly Business-Demography Measurements	12
8	State Laws	13
9	Joint Reference-Markup Block	13
10	Inflation Equation	14

11 Two-Scale Approximation	14
12 Model Families	15
12.1 Cell 1: Quantitative HSA Hierarchy	15
12.2 Cells 2–9: Robustness and Transportability Hierarchy	16
13 Full-Joint and Modular-Cut Designs	17
13.1 Modular Cut	17
13.2 Full Joint	17
14 Posterior Simulation	18
15 Priors	18
16 Identification Diagnostics	19
16.1 Competition-Cycle Measurement	19
16.2 Identification of θ_0 and γ	20
16.3 Persistent Inflation Disturbance	20
16.4 Low-Frequency Inflation Confounding	21
17 Measurement-Error and Data-Alignment Checks	21
17.1 Markup Measurement Error	21
17.2 Shared Price-Index Construction	21
17.3 Market-Universe Alignment	21
18 Model Evidence and Predictive Evaluation	22
18.1 Bayes Factors	22
18.2 Posterior Predictive Score	22
18.3 Posterior Predictive Checks	23
19 Cross-Cell Interpretation	23
20 Required Robustness Exercises	24

21 Required Simulation Validation	25
22 Reporting Plan	25
22.1 Main Competition-State Figure	25
22.2 Main Cell-1 Figure	25
22.3 Nine-Cell Coefficient Figure	26
22.4 Evidence Table	26
23 Pre-Estimation Freeze	26
24 Final Interpretation Boundary	27

1 Objective and Scope

The design studies whether product-market competition changes inflation dynamics through two empirically distinct channels:

1. a competition-dependent marginal-cost or slack slope; and
2. a short-run market-participation channel whose loading may itself vary with the competitive environment.

The analysis is deliberately semi-structural. The business-demography system is used to construct a latent empirical competition coordinate, while HSA theory supplies a local cross-equation restriction only in the empirical cell for which the inflation and marginal-cost measures are sufficiently close to the producer-pricing objects in the theory.

The empirical analysis is organized around a pre-declared 3×3 matrix of nine data cells. The nine cells are not treated symmetrically in structural interpretation. One cell is the primary quantitative HSA cell; the remaining eight are mechanism-robustness or transportability exercises.

The design distinguishes:

- quantitative HSA consistency;
- robustness of the competition mechanisms; and
- transportability of those mechanisms across alternative inflation and slack measures.

No result from a secondary cell is used to redefine the primary HSA restriction *ex post*.

2 Research Questions

2.1 RQ1: Competition-Dependent Slope

Does the slow competitive environment change the response of inflation to marginal cost or macroeconomic slack?

The empirical object is

$$\kappa_1 = \frac{\partial}{\partial \bar{q}} \left(\frac{\partial \pi}{\partial x} \right) \bigg|_{\bar{q}=0}. \quad (1)$$

2.2 RQ2: Short-Run Participation Channel

Conditional on the maintained timing, measurement, and shock-orthogonality assumptions, is the short-run competition cycle associated with inflation through a distinct participation-channel loading?

The empirical object is

$$\theta_0 = - \left. \frac{\partial \pi}{\partial \hat{q}} \right|_{\bar{q}=0}. \quad (2)$$

This coefficient is not interpreted as a nonparametrically identified causal effect.

2.3 RQ3: Local HSA Restriction

In the theory-near PPI/inverse-markup cell, are the slope and participation channels quantitatively linked as HSA predicts?

The local restriction is

$$\boxed{\kappa_1 = \zeta_0 \theta_0.} \quad (3)$$

For the unrestricted model define

$$\delta_{HSA} = \kappa_1 - \zeta_0 \theta_0. \quad (4)$$

This restriction is tested only in the primary quantitative HSA cell.

2.4 RQ4: State Dependence of the Participation Loading

Does the participation-channel loading vary with the slow competitive environment?

Write

$$\theta(\bar{q}) = \theta_0 + \gamma \bar{q} + O(\bar{q}^2). \quad (5)$$

The primary empirical interpretation of γ is the local state dependence of the empirical participation-channel loading.

A direct interpretation of γ as a derivative of structural pass-through additionally requires that the mapping between the theoretical effective-variety coordinate and the empirical competition coordinate have locally constant scale at the reference environment.

Thus $\gamma = 0$ means local constancy of the empirical participation loading. It is interpreted as locally constant structural pass-through only under this additional proxy-mapping condition.

2.5 RQ5: Transportability

Are the signs, magnitudes, and state dependence of the competition mechanisms stable across alternative inflation indexes and alternative marginal-cost/slack measures?

3 Nine Pre-Declared Empirical Cells

Quarterly inflation is considered for three price indexes:

1. PPI;
2. headline CPI;
3. core CPI.

The activity or marginal-cost block is considered under three measures:

1. inverse markup;
2. Beveridge–Nelson (BN) output gap;
3. negative unemployment gap.

The resulting design is shown in table 1.

Table 1: Nine pre-declared empirical cells

Inflation / expectation block	Inverse markup	BN output gap	Negative unemployment gap
PPI / GDP-deflator expectation	Cell 1	Cell 2	Cell 3
CPI / CPI expectation	Cell 4	Cell 5	Cell 6
Core CPI / CPI expectation	Cell 7	Cell 8	Cell 9

The nine cells are frozen before empirical comparison.

3.1 Tier 1: Quantitative HSA Cell

Cell 1: $\text{PPI} \times \text{inverse markup}$. This is the unique primary quantitative HSA cell.

PPI is the closest observed inflation measure to the producer-pricing equation underlying the HSA restriction, while inverse markup provides the closest direct empirical mapping to real marginal cost.

Only this cell receives the quantitative restriction in eq. (3).

3.2 Tier 2: Mechanism Robustness

The following cells move one major dimension away from the theory-near benchmark:

- Cell 2: $\text{PPI} \times \text{BN output gap}$;
- Cell 3: $\text{PPI} \times \text{negative unemployment gap}$;
- Cell 4: $\text{CPI} \times \text{inverse markup}$.

These cells ask whether the estimated competition mechanisms survive alternative measures of marginal cost/slack or an alternative inflation index.

They do not receive the quantitative HSA equality.

3.3 Tier 3: Macro Transportability

The remaining cells are:

- Cell 5: $\text{CPI} \times \text{BN output gap}$;
- Cell 6: $\text{CPI} \times \text{negative unemployment gap}$;
- Cell 7: $\text{core CPI} \times \text{inverse markup}$;
- Cell 8: $\text{core CPI} \times \text{BN output gap}$;
- Cell 9: $\text{core CPI} \times \text{negative unemployment gap}$.

These cells study whether the competition mechanisms transport to conventional or underlying consumer-price Phillips curves.

No structural HSA equality is imposed in these cells.

4 Inflation and Expectation Measures

For price index $j \in \{PPI, CPI, Core\}$, quarterly annualized inflation is

$$\pi_t^j = 400 \left(\log P_t^j - \log P_{t-1}^j \right). \quad (6)$$

All expectation regressors are one-quarter-ahead forecasts whenever available and are expressed in corresponding quarterly annualized units.

4.1 PPI Cells

The preferred conditioning variable is a one-quarter-ahead PPI expectation.

If a sufficiently long PPI expectation series is unavailable, a one-quarter-ahead GDP-deflator expectation is used as an observed conditioning proxy. It is not relabeled as a PPI expectation.

The baseline therefore conditions on

$$g_t^{GDP} \quad (7)$$

as a nuisance expectation regressor.

A mandatory robustness exercise introduces an explicit PPI–GDP expectation bridge whenever sufficient information exists to identify it.

4.2 CPI Cells

Headline CPI inflation is paired with a one-quarter-ahead CPI expectation.

4.3 Core CPI Cells

If a direct core-CPI expectation is unavailable, headline CPI expectation is used as a conditioning proxy. It is explicitly labeled as a headline-CPI expectation proxy rather than an expected core-inflation measure.

A headline-to-core expectation bridge is a robustness exercise when sufficient data are available.

5 Marginal-Cost and Slack Measures

5.1 Inverse Markup

Let

$$\mu_t \tag{8}$$

denote the observed markup. Real marginal cost is $1/\mu_t$.

No sample-mean normalization is used in the structural inverse-markup cells.

Instead, define the marginal-cost deviation relative to the same reference environment used for the HSA elasticity:

$$\boxed{x_t^{IM} = \log(1/\mu_t) - \log(1/\mu_{\text{ref}}^f) = \log\left(\frac{\mu_{\text{ref}}^f}{\mu_t}\right)}. \tag{9}$$

Thus $x_t^{IM} = 0$ has an economic interpretation: observed real marginal cost equals its reference flexible-price/steady-state value.

This eliminates dependence on an arbitrary sample-mean centering constant.

5.2 BN Output Gap

Let

$$x_t^{BN} \tag{10}$$

denote the pre-declared BN output gap.

The zero of the measure already has the interpretation of output being at its estimated permanent component, so no additional centering is applied.

5.3 Negative Unemployment Gap

Let

$$u_t^{gap} = u_t - u_t^*. \quad (11)$$

Define

$$x_t^U = -u_t^{gap}. \quad (12)$$

The sign convention makes larger values correspond to a tighter labor market.

No additional normalization is applied.

The unemployment and BN measures are not interpreted as being in raw log marginal-cost units. Therefore their coefficients are not directly subject to the HSA quantitative equality.

6 HSA Theory and Local Restriction

Let

$$n = \log N \quad (13)$$

denote the theoretical effective number of active firms or varieties.

Under the maintained Rotemberg-HSA producer-pricing environment,

$$\kappa^{th}(n) = \frac{\zeta(n) - 1}{\chi}, \quad (14)$$

and

$$\Theta^{th}(n) = \frac{1}{\chi} \frac{1 - \rho(n)}{\rho(n)}. \quad (15)$$

The HSA identity is

$$\frac{d\kappa^{th}(n)}{dn} = \zeta(n)\Theta^{th}(n). \quad (16)$$

Locally relate the empirical competition coordinate to the theoretical coordinate by

$$q = b_N(n - n_0), \quad b_N > 0. \quad (17)$$

Under a common local rescaling,

$$\frac{d\kappa(q)}{dq} = \zeta(q)\theta(q). \quad (18)$$

At the reference environment

$$\bar{q} = 0, \quad \hat{q} = 0, \quad (19)$$

the primary quantitative HSA restriction is eq. (3).

This is a pointwise local restriction, not a globally exact coefficient function.

It is imposed only in Cell 1.

7 Competition Measurement System

The latent empirical competition coordinate is

$$q_t = \bar{q}_t + \hat{q}_t, \quad (20)$$

where:

- $\bar{q}_t$ is a slowly moving competitive environment;
- $\hat{q}_t$ is a stationary competition cycle.

The state decomposition is a measurement and regularization device rather than an HSA law.

7.1 Annual Firm Counts

Let

$$q_y^A \quad (21)$$

denote the centered annual firm-count coordinate.

For a point-in-time annual observation at quarter t_y ,

$$q_y^A = \bar{q}_{t_y} + \hat{q}_{t_y} + \nu_y^N, \quad \nu_y^N \sim \mathcal{N}(0, \sigma_N^2). \quad (22)$$

The loading of one defines the scale of the empirical competition coordinate.

The annual firm count is treated as the defining empirical business-demography coordinate, not as an exact observation of theoretical effective varieties.

No additional latent time-varying proxy wedge is included in the baseline unless more than one independently informative annual competition measure is available, because a single annual series cannot separately identify the common competition state and a flexible persistent proxy wedge.

Instead, the design uses:

- sensitivity to σ_N ;

- alternative annual competition proxies;
- market-universe checks;
- a multiple-proxy measurement model if multiple credible annual measures become available.

7.2 Quarterly Business-Demography Measurements

Let

$$e_{jt}, \quad j = 1, \dots, J_E, \quad (23)$$

denote economically comparable quarterly business-demography indicators.

The baseline measurement equation is

$$e_{jt} = \bar{e}_{jt} + b_j \hat{q}_t + \xi_{jt}^E, \quad \xi_{jt}^E \sim \mathcal{N}(0, \sigma_{E,j}^2), \quad (24)$$

where each series receives its own slow component $\bar{e}_{jt}$.

The slow establishment or business-demography components are not tied to $\bar{q}_t$.

The loading b_j is estimated.

Only quarterly indicators that provide genuinely distinct measurement information are included. Additional series are not added merely to increase the number of measurement rows.

If only one credible quarterly establishment series is available, the system reduces to

$$e_t = \bar{e}_t + b_E \hat{q}_t + \xi_t^E. \quad (25)$$

A mandatory conservative sensitivity allows each quarterly indicator to contain an additional idiosyncratic stationary cycle:

$$e_{jt} = \bar{e}_{jt} + b_j \hat{q}_t + r_{jt}^E + \xi_{jt}^E, \quad (26)$$

$$r_{jt}^E = \rho_{E,j} r_{j,t-1}^E + \omega_{jt}^E, \quad |\rho_{E,j}| < 1. \quad (27)$$

If the common loading or the posterior sharpening of $\hat{q}_t$ disappears under this specification, the quarterly data are not considered sufficiently informative about the common competition cycle.

8 State Laws

The baseline competition-state laws are

$$\bar{q}_t = d_q + \bar{q}_{t-1} + \eta_t^q, \quad \eta_t^q \sim \mathcal{N}(0, \sigma_q^2), \quad (28)$$

$$\hat{q}_t = \phi_q \hat{q}_{t-1} + u_t^q, \quad u_t^q \sim \mathcal{N}(0, \sigma_q^2), \quad |\phi_q| < 1. \quad (29)$$

Each quarterly business-demography slow component obeys a pre-declared smooth law, with a random walk with drift as the baseline:

$$\bar{e}_{jt} = d_{e,j} + \bar{e}_{j,t-1} + \eta_{jt}^e. \quad (30)$$

Alternative smooth laws for $\bar{q}_t$, $\hat{q}_t$, and $\bar{e}_{jt}$ are identification sensitivities rather than alternative economic theories.

9 Joint Reference-Markup Block

The flexible-price/steady-state reference markup is estimated jointly rather than fixed in a preliminary calibration.

Parameterize

$$h_\mu = \log(\mu_{\text{ref}}^f - 1), \quad (31)$$

so that

$$\mu_{\text{ref}}^f = 1 + \exp(h_\mu) > 1. \quad (32)$$

The associated HSA elasticity is

$$\boxed{\zeta_0 = \frac{\mu_{\text{ref}}^f}{\mu_{\text{ref}}^f - 1} = 1 + \exp(-h_\mu).} \quad (33)$$

This parameterization guarantees the economically required domain.

A baseline low-frequency markup block is

$$\log \mu_t^{\text{obs}} = \log \mu_{\text{ref}}^f + \lambda_\mu \bar{q}_t + r_t^\mu + v_t^\mu, \quad v_t^\mu \sim \mathcal{N}(0, \sigma_{\mu,v}^2), \quad (34)$$

with

$$r_t^\mu = \rho_\mu r_{t-1}^\mu + \omega_t^\mu, \quad |\rho_\mu| < 1, \quad \omega_t^\mu \sim \mathcal{N}(0, \sigma_{\mu,r}^2). \quad (35)$$

This equation is an empirical bridge between observed markups and the reference flexible-price

markup. It is not treated as an identity equating observed cyclical markups with their flexible-price counterparts.

For every posterior draw,

$$x_t^{IM} = \log \left(\frac{\mu_{\text{ref}}^f}{\mu_t^{obs}} \right), \quad (36)$$

and, in Cell 1,

$$\kappa_1 = \zeta_0 \theta_0 \quad (37)$$

is imposed draw by draw.

Thus reference-markup uncertainty is propagated directly into both the marginal-cost gap and the HSA restriction.

10 Inflation Equation

For each empirical cell c , the encompassing local equation is

$$\boxed{\begin{aligned} \pi_{c,t} = & a_c + \beta_{b,c} \pi_{c,t-1} + \beta_{f,c} f_{c,t}^{cond} + \kappa_{0,c} x_{c,t} + \kappa_{1,c} \bar{q}_t x_{c,t} \\ & - \theta_{0,c} \hat{q}_t - \gamma_c \bar{q}_t \hat{q}_t + \varepsilon_{c,t}^\pi, \end{aligned}} \quad (38)$$

with

$$\varepsilon_{c,t}^\pi \sim \mathcal{N}(0, \sigma_{\pi,c}^2) \quad (39)$$

in the baseline.

The lagged- and expected-inflation coefficients are estimated separately.

No sum-to-one restriction is imposed.

The coefficient interpretations are

$$\kappa_{1,c} = \frac{\partial}{\partial \bar{q}} \left(\frac{\partial \pi_c}{\partial x_c} \right), \quad (40)$$

$$\theta_{0,c} = - \left. \frac{\partial \pi_c}{\partial \hat{q}} \right|_{\bar{q}=0}, \quad (41)$$

and

$$\gamma_c = - \frac{\partial^2 \pi_c}{\partial \bar{q} \partial \hat{q}}. \quad (42)$$

11 Two-Scale Approximation

The baseline is a two-scale varying-coefficient approximation.

The slow state $\bar{q}_t$ indexes the competitive environment. Conditional on that environment, x_t and $\hat{q}_t$

are treated as fast cyclical variables.

Thus

$$\kappa(\bar{q}_t) \simeq \kappa_0 + \kappa_1 \bar{q}_t, \quad (43)$$

and

$$\theta(\bar{q}_t) \simeq \theta_0 + \gamma \bar{q}_t. \quad (44)$$

The retained terms are therefore

$$\kappa_0 x_t + \kappa_1 \bar{q}_t x_t - \theta_0 \hat{q}_t - \gamma \bar{q}_t \hat{q}_t. \quad (45)$$

The baseline exclusion of

$$\hat{q}_t x_t \quad (46)$$

is not an HSA zero restriction.

It is the maintained approximation that fast-by-fast interactions are excluded while slow-environment variation in fast-variable coefficients is retained.

A mandatory sensitivity adds

$$\lambda_{qx} \hat{q}_t x_t \quad (47)$$

symmetrically to the relevant restricted and unrestricted models.

A stronger sensitivity imposes

$$\lambda_{qx} = \kappa_1, \quad (48)$$

corresponding to a first-order single-coordinate expansion in total

$$q_t = \bar{q}_t + \hat{q}_t. \quad (49)$$

If the principal result changes materially under either specification, the conclusion is reported as dependent on the two-scale approximation.

12 Model Families

12.1 Cell 1: Quantitative HSA Hierarchy

Cell 1 uses four nested models.

M0: No competition channel.

$$\kappa_1 = \theta_0 = \gamma = 0. \quad (50)$$

M1: Local constant-participation HSA.

$$\kappa_1 = \zeta_0 \theta_0, \quad \gamma = 0. \quad (51)$$

M2: Local HSA.

$$\kappa_1 = \zeta_0 \theta_0, \quad \theta_0, \gamma \text{ free.} \quad (52)$$

M3: Local unrestricted competition.

$$\kappa_1, \theta_0, \gamma \text{ free.} \quad (53)$$

The primary HSA residual under M3 is eq. (4).

12.2 Cells 2–9: Robustness and Transportability Hierarchy

The HSA equality is not imposed.

Each cell uses:

R0: No competition channel.

$$\kappa_1 = \theta_0 = \gamma = 0. \quad (54)$$

R1: Competition with locally constant participation loading.

$$\kappa_1, \theta_0 \text{ free,} \quad \gamma = 0. \quad (55)$$

R2: Unrestricted local competition.

$$\kappa_1, \theta_0, \gamma \text{ free.} \quad (56)$$

Thus the secondary cells answer two clean questions:

1. do competition-related terms improve the inflation model?;
2. if so, is state dependence of the participation loading empirically required?

No δ_{HSA} is interpreted structurally outside Cell 1.

13 Full-Joint and Modular-Cut Designs

Let C collect annual and quarterly business-demography measurements.

Let S collect competition states and their measurement parameters.

For inverse-markup cells, let ψ_μ collect reference-markup parameters and latent markup components.

13.1 Modular Cut

The primary measurement-supported cut for inverse-markup cells is

$$p_{cut}(S, \psi_\mu, \beta_c) = p(S | C) p(\psi_\mu | \mu^{obs}, S) p(\beta_c | \pi_c, X_c, S, \psi_\mu). \quad (57)$$

This ordering has a deliberate interpretation:

1. business-demography data determine competition states;
2. markup data determine the reference-markup block conditional on those states;
3. inflation determines inflation-equation parameters;
4. inflation does not reweight competition states or the reference-markup calibration.

For BN-output-gap and unemployment-gap cells the middle markup module is absent:

$$p_{cut}(S, \beta_c) = p(S | C) p(\beta_c | \pi_c, X_c, S). \quad (58)$$

This modular design gives all nine cells the same business-demography-supported competition-state benchmark.

13.2 Full Joint

For inverse-markup cells,

$$\begin{aligned} p_{joint}(S, \psi_\mu, \beta_c | C, \mu^{obs}, \pi_c, X_c) \\ \propto p(C | S) p(\mu^{obs} | S, \psi_\mu) p(\pi_c | S, \psi_\mu, X_c, \beta_c) p(S) p(\psi_\mu) p(\beta_c). \end{aligned} \quad (59)$$

For gap cells, the markup block is omitted.

Full-joint inference is a coherent semi-structural pooling estimator.

It is not an endogeneity correction.

Cut-joint differences are treated as information-flow and modular-conflict diagnostics.

14 Posterior Simulation

The bilinear term

$$-\gamma \bar{q}_t \hat{q}_t \tag{60}$$

makes the joint state posterior non-Gaussian in the full state vector.

Conditional on either slow or fast state blocks, however, the other block has a linear-Gaussian full conditional.

Posterior inference therefore uses an

$$\boxed{\text{FFBS-within-Gibbs sampler}} \tag{61}$$

rather than describing the complete posterior sampler as an exact FFBS algorithm.

Each state-block FFBS step is an exact draw from its corresponding conditional smoothing distribution.

Repeated Gibbs iterations target the joint posterior.

A typical iteration is:

1. draw $\hat{q}_{1:T}$ and the quarterly business-demography slow states conditional on $\bar{q}_{1:T}$;
2. draw $\bar{q}_{1:T}$ conditional on $\hat{q}_{1:T}$;
3. update transition and measurement parameters;
4. update quarterly-measurement loadings and variances;
5. in inverse-markup cells, update the reference-markup block;
6. update inflation-equation parameters;
7. repeat the state blocks before storing a draw if cross-block autocorrelation is high.

Gaussian conditional updates are used where available.

The reference-markup parameter h_μ , and any other parameter entering the likelihood nonlinearly, is updated with a valid nonconjugate step such as slice sampling or Metropolis–Hastings.

When $\gamma = 0$, a one-block multivariate FFBS implementation is used as a mandatory computational validation benchmark.

15 Priors

All priors are proper.

Near-improper inverse-gamma variance priors are excluded.

The prior design focuses on:

- the relative innovation scales of $\bar{q}_t$ and $\hat{q}_t$;
- quarterly-measurement trend smoothness;
- the loading parameters b_j ;
- the annual measurement variance;
- inflation-equation coefficient scales;
- the reference-markup parameter h_μ .

The prior on

$$h_\mu = \log(\mu_{\text{ref}}^f - 1) \quad (62)$$

is based on economically plausible markup magnitudes or external evidence and is not tuned using realized inflation outcomes.

Prior predictive checks examine the implied distributions of

$$\kappa_1 \bar{q}_t x_t, \quad -\theta_0 \hat{q}_t, \quad -\gamma \bar{q}_t \hat{q}_t, \quad (63)$$

and, in Cell 1,

$$\delta_{HSA}. \quad (64)$$

Bayes-factor prior scales are frozen before the model comparisons and are varied over pre-declared economically plausible ranges.

16 Identification Diagnostics

Identification diagnostics are co-primary with coefficient estimates.

16.1 Competition-Cycle Measurement

Compare:

N. Annual firms only.

C. Annual firms plus quarterly business-demography measurements.

J. Full-joint cell-specific model.

J-Q. Full-joint model with the quarterly business-demography signal removed.

The central measurement diagnostic is

$$C \text{ versus } N \tag{65}$$

for posterior uncertainty about

$$\hat{q}_t. \tag{66}$$

The quarterly data are considered informative only if they materially sharpen the competition cycle and the result survives:

- alternative quarterly trend laws;
- alternative competition-cycle laws;
- idiosyncratic quarterly cycles;
- alternative quarterly indicators.

16.2 Identification of θ_0 and γ

For every main cell report:

- posterior variation in $\bar{q}_t$;
- posterior variation in $\hat{q}_t$;
- correlation between $\hat{q}_t$ and $\bar{q}_t\hat{q}_t$;
- standardized-design condition number;
- smallest eigenvalue of the regressor Gram matrix;
- auxiliary R^2 and VIF for $\bar{q}_t\hat{q}_t$;
- posterior correlation between θ_0 and γ ;
- prior-to-posterior contraction;
- cut-joint movement in the coefficients.

If γ is not learned from the likelihood, posterior concentration around zero or a Bayes factor favoring $\gamma = 0$ is not interpreted as evidence for local constancy.

16.3 Persistent Inflation Disturbance

A mandatory robustness replaces

$$\varepsilon_t^\pi \sim iid \tag{67}$$

with

$$\varepsilon_t^\pi = \rho_\pi \varepsilon_{t-1}^\pi + v_t^\pi, \quad |\rho_\pi| < 1. \quad (68)$$

A competition-cycle loading is not considered robust if it collapses once modest residual inflation persistence is admitted.

16.4 Low-Frequency Inflation Confounding

A common low-frequency inflation nuisance component is added symmetrically across competing models within each cell.

If κ_1 changes materially, the result is described as sensitive to low-frequency inflation specification.

17 Measurement-Error and Data-Alignment Checks

17.1 Markup Measurement Error

Inverse-markup results are conditional on the observed markup unless an identified errors-in-variables structure is available.

A valid measurement-error model requires additional information, such as:

- externally estimated measurement-error variance;
- independently constructed markup measures;
- an external instrument.

A measurement equation alone is not treated as identification.

17.2 Shared Price-Index Construction

If the markup measure mechanically contains the same price index used to construct inflation, a shared-measurement-error sensitivity is mandatory.

Mechanical covariance is not interpreted as marginal-cost transmission.

17.3 Market-Universe Alignment

PPI, CPI/core CPI, markup measures, firm counts, and quarterly business-demography indicators must refer to economically compatible market universes.

Cell 7, core CPI $\times$ inverse markup, receives particular scrutiny because a broad markup measure may not map cleanly into the core-CPI market universe.

If the alignment is materially poor, the cell is reported as non-interpretable rather than forced into the nine-cell comparison.

18 Model Evidence and Predictive Evaluation

Model comparison uses two distinct objects.

18.1 Bayes Factors

Bayes factors measure integrated model evidence, including fit, parameter uncertainty, prior mass, and model complexity.

They are therefore not described as pure fit measures.

Bayes factors are compared only between models estimated on the same empirical cell.

They are not compared numerically across different dependent variables or different activity measures.

For Cell 1 the main comparisons are:

$$M3 \text{ vs } M0, \tag{69}$$

$$M2 \text{ vs } M3, \tag{70}$$

and

$$M1 \text{ vs } M2. \tag{71}$$

For Cells 2–9 the main comparisons are:

$$R2 \text{ vs } R0, \tag{72}$$

and

$$R1 \text{ vs } R2. \tag{73}$$

Formal Bayes factors are based on the full generative model.

No mechanically comparable Bayes factor is assigned to the modular cut distribution.

18.2 Posterior Predictive Score

Predictive evaluation uses time-series-respecting posterior predictive densities.

The preferred criterion is recursive one-step-ahead or leave-future-out log predictive score:

$$LPS_M = \sum_{t \in \mathcal{H}} \log p(\pi_t \mid \mathcal{I}_{t-1}, M). \tag{74}$$

Full-sample smoothed latent states are never used to claim forecast performance.

Where contemporaneous activity or markup data are unavailable at the genuine forecast origin, a separate conditional-nowcast exercise is reported.

Predictive-score differences are interpreted within each cell relative to its no-competition benchmark.

18.3 Posterior Predictive Checks

Posterior predictive checks are used as absolute-fit diagnostics for:

- inflation persistence;
- residual variance;
- business-demography measurement fit;
- markup-block fit;
- the magnitude of competition contributions.

Thus the evidence hierarchy is

$$\boxed{\text{Identification} \rightarrow \text{Bayes factor} + \text{predictive score} \rightarrow \text{posterior predictive fit.}} \quad (75)$$

A favorable Bayes factor cannot rescue a model whose key parameter is weakly identified.

19 Cross-Cell Interpretation

The nine cells are not nine independent attempts to find statistical significance.

They are a fixed robustness architecture.

The interpretation proceeds in the following order.

Gate 1: Competition measurement. Does the quarterly business-demography system identify a meaningful $\hat{q}_t$ beyond annual firm counts and state-law regularization?

Gate 2: Primary HSA cell. In Cell 1:

- is θ_0 empirically learned?;
- is κ_1 empirically learned?;
- is δ_{HSA} close to zero?;
- does Local HSA receive competitive integrated evidence relative to Local U?;

- do the results survive persistent inflation errors, alternative competition state laws, and the fast-by-fast interaction sensitivity?

Gate 3: Marginal-cost/slack robustness. Do Cells 2 and 3 produce competition-dependent Phillips-curve slopes and participation loadings with economically compatible signs?

No quantitative HSA equality is required.

Gate 4: Inflation-index transportability. Do Cells 4–9 show that the mechanism survives movement from producer inflation to headline and core consumer inflation?

Gate 5: State dependence. Only if γ is empirically learned is variation in the participation loading interpreted substantively.

20 Required Robustness Exercises

The following are mandatory:

1. annual-firm-only versus quarterly-augmented competition measurement;
2. alternative quarterly business-demography trends;
3. idiosyncratic quarterly business-demography cycles;
4. alternative AR(1)/AR(2) competition-cycle laws;
5. alternative slow competition laws;
6. persistent AR(1) inflation disturbance;
7. low-frequency inflation nuisance component;
8. fast-by-fast interaction $\hat{q}_t x_t$;
9. stronger single-coordinate restriction $\lambda_{qx} = \kappa_1$;
10. markup measurement-error sensitivity where identified;
11. shared-price-index measurement-error check;
12. alternative annual competition proxies;
13. wider annual-measurement variance;
14. expectation-index bridge checks;
15. cut versus full-joint information-flow diagnostics;

16. market-universe alignment;
17. prior sensitivity for all Bayes-factor comparisons;
18. local-approximation range diagnostics.

21 Required Simulation Validation

Before empirical model ranking, simulation exercises must verify:

1. agreement between one-block and two-block state samplers when $\gamma = 0$;
2. recovery of θ_0 , γ , and κ_1 under known parameter values;
3. deterioration of γ identification when variation in $\bar{q}_t$ is small;
4. the ability of the C-versus-N diagnostic to reveal weak quarterly information;
5. behavior when quarterly indicators contain idiosyncratic cycles;
6. false loading on $\hat{q}_t$ when inflation disturbances are persistent;
7. rejection of the HSA-restricted model when data are generated with economically important $\delta_{HSA} \neq 0$;
8. cut-joint state shifts under a correctly specified joint model;
9. recovery of the reference markup and ζ_0 ;
10. RBPF likelihood validation against exact Kalman-filter likelihoods in linear special cases.

22 Reporting Plan

22.1 Main Competition-State Figure

Report

$$\bar{q}_t, \quad \hat{q}_t, \quad q_t \tag{76}$$

under the business-demography cut and full joint.

Show the C-versus-N uncertainty reduction for $\hat{q}_t$.

22.2 Main Cell-1 Figure

Report posterior distributions of

$$\kappa_1, \quad \theta_0, \quad \gamma, \quad \delta_{HSA}, \quad \mu_{\text{ref}}^f, \quad \zeta_0. \tag{77}$$

22.3 Nine-Cell Coefficient Figure

Report posterior distributions or credible intervals for

$$\kappa_1, \quad \theta_0, \quad \gamma \tag{78}$$

for all nine cells.

The figure visually distinguishes:

- Tier 1 quantitative HSA;
- Tier 2 mechanism robustness;
- Tier 3 transportability.

No structural HSA residual is plotted for Cells 2–9.

22.4 Evidence Table

For each cell report:

- Bayes-factor comparisons within that cell;
- recursive/LFO predictive-score differences relative to the no-competition model;
- cut and full-joint coefficient summaries;
- identification diagnostics;
- persistent-error robustness;
- convergence and Monte Carlo diagnostics.

23 Pre-Estimation Freeze

Before production estimation, freeze:

1. all nine empirical cells;
2. Tier 1, Tier 2, and Tier 3 classification;
3. all inflation definitions and forecast horizons;
4. all activity/slack definitions;
5. markup construction;
6. reference-markup model;

7. business-demography sources;
8. annual and quarterly timing conventions;
9. competition-state laws;
10. priors;
11. model families within each cell;
12. modular-cut ordering;
13. full-joint specifications;
14. identification gates;
15. predictive information sets;
16. Bayes-factor prior sensitivity grid;
17. all mandatory robustness exercises.

No cell is added, deleted, or promoted based on the realized coefficient signs or model rankings, except that a cell may be labeled non-interpretable if a pre-declared market-universe or data-quality condition fails.

24 Final Interpretation Boundary

A favorable result in the primary cell supports the following deliberately conditional statement:

Annual firm counts and quarterly business-demography measurements provide an empirically informative latent competition coordinate under the maintained mixed-frequency state-space decomposition. Conditional on the timing, measurement, reference-markup, and shock-orthogonality assumptions, competition changes the marginal-cost slope and is associated with a short-run participation-channel loading. In the PPI/inverse-markup cell, the two channels are locally quantitatively consistent with the HSA restriction after uncertainty in the reference flexible-price markup and the implied demand elasticity is propagated through the joint model.

The secondary cells establish robustness or transportability only.

The design does not by itself establish:

- that observed firm counts equal theoretical effective varieties;
- that the competition state laws are structural;
- that θ_0 is a nonparametrically identified causal effect;

- that γ is a structural pass-through derivative without the additional local proxy-mapping condition;
- that unemployment or output gaps are in the same units as theoretical marginal cost;
- that CPI or core CPI obey the producer-pricing HSA equation;
- that full-joint estimation solves endogenous entry.

Those stronger claims require additional measurement structure or independently identified sources of variation.